

Supplement S3: Reproducibility

The directory `repro/` is a clean publication reproducibility package. From that directory run

```
python3 scripts/build_all.py generated
```

It requires Python 3 and pdfLaTeX/TikZ, uses no network and no randomness, and includes the semantic reference implementation used for the bounded validation. The scripts regenerate the all-64 table, the $k = 1$ table, validation JSON, editable vector figure sources, SVG figures, and PDF figures. The bounded validation products provide independent finite verification and reproducibility evidence only; they are not premises of the symbolic all- k proofs.